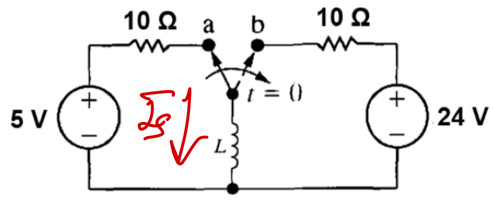
Tutorial-4 Quiz Solutions

Q1. You are analyzing a fuel injector solenoid. It can be modeled as a series R-L circuit (R = 10 Ω, L=100 mH). Instead of a simple on/off switch, the circuit is controlled by a 2-Position Switch, as shown in the figure.



The switch has been in Position A for a long time (Steady State 1). At t=0, the switch instantly flips to Position B.

A. Calculate the current flowing just before the switch flips, $i(0^-)$. What is the current $i(0^+)$ the exact instant the 24 V source is connected? Why doesn't the current jump to 2.4 A immediately?

B. At $t=0^+$, calculate the voltage across the inductor $v_L(0^+)$ at this exact moment. How can the inductor voltage be *almost* as high as the source?

C. Write the mathematical expression for i(t) for t > 0. *(Hint: Use the General Solution* $i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$*).*

D. However, the fuel injector is a mechanical valve held shut by a stiff spring. The magnetic force is only strong enough to pull the valve open once the current exceeds 1.5A. Use your equation to calculate the exact time t_{open} when the current crosses the 1.5A threshold.

E. The engine control unit (ECU) needs this injector to open within 3 ms of the command signal to ensure fuel enters the cylinder at the correct moment for combustion. Does your circuit meet this specification, or is the inductor too slow?

A. For $t<0$, switch is at a.

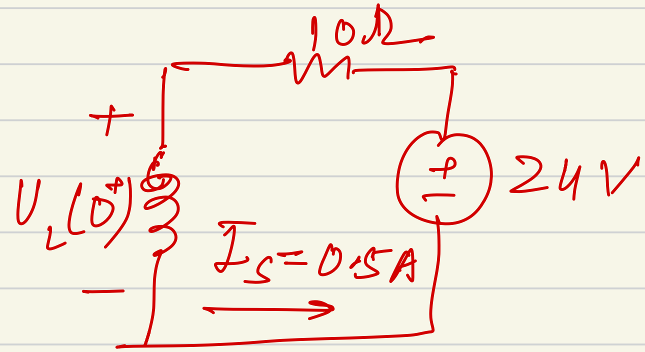
In steady state, $L \to$ S.C.

$$i(0^-) = I_s = \frac{5}{10} = 0.5\,A$$

at $t=0$, switch flips to b.

$i(0^+) = 0.5\,A$ (due to current intertia of the inductor)

B.



$$V_L(0^+) + 10 \times 0.5 - 24 = 0$$

$$V_L(0^+) = 19\,V$$

Because $i(0^-)$ dictates $V_L(0^+)$.

C. $i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$, $\tau = \frac{L}{R} = \frac{0.1}{10} = 0.01s = 10ms$

$i(\infty)$ can be calculated when $L \to S.C$ in 24 V loop.

$i(\infty) = \frac{24}{10} = 2.4A$

$i(0) = 0.5A$, $\boxed{i(t) = 2.4 - 1.9e^{-t\times 100}}$

D. at $t=0$, $i(0) = 0.5A$

at $t = t_{open}$. $i(t_{open}) \geq 1.5A$

$1.5 = 2.4 - 1.9e^{-t/100} \Rightarrow -0.9 = -1.9e^{-t/0.01}$

$\Rightarrow \frac{0.9}{1.9} = e^{-t/0.01}$

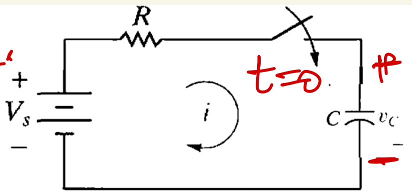
$\Rightarrow \ln\left(\frac{1.9}{0.9}\right) = \frac{t}{0.01} \Rightarrow t = 0.01 \times \ln\left(\frac{1.9}{0.9}\right)$

$\simeq 7.8$ ms

E. $t_{open} > 3ms$, your designed circuit time constant is more than required.

Q2. You are designing the Burst Mode for a professional camera flash. The given circuit models the phases: phase2.

- Phase 1 (t<0): When the flash fires, the voltage drops rapidly. However, the Xenon tube stops conducting (extinguishes) when the voltage drops to 50 V. It doesn't go to zero! ($v_c(0) = 50\ V$.)
- Phase 2 (t=0+): Immediately after the flash cuts off, the charging circuit activates to refill the capacitor.



$V_s > V_c(0)$

A photographer takes a picture at t < 0. The flash fires and extinguishes, leaving the capacitor with a Residual Voltage of 50 V.

A. Calculate the initial charging current i(0+) the moment the recharge begins. Does having this 50 V head start make the initial current higher or lower compared to charging from 0V?

B. Write the mathematical expression for the voltage v_c(t) for t > 0. *(Hint: Use the General Solution* $v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$*).*

C. The photographer is shooting a fast sports event and presses the shutter button again just 0.8 s after the first shot. Calculate the voltage across the capacitor at exactly t=0.8 s.

D. For a "Good Exposure," the flash needs to deliver at least 15 J of energy. Did the capacitor accumulate enough energy by 0.8s to take a good photo, or will the second picture be underexposed (dark)?

$V_c(0) = 50V.$ $\quad V_s = 300V,\ C = 1000\mu F$
$R = 1k\Omega$

at $t=0$, when switch closes $\rightarrow$ $V_c(0^+) = 50V$ because your cap is S.C.

A. $i(0^+) = \dfrac{V_s - V_c(0)}{R} = \dfrac{300-50}{1K} = 250\,mA = 0.25\,A$

if $V_c(0) = 0$, $\quad i(0^+) = \dfrac{V_s}{R}$, it is higher than i (when $V_c(0)=50V$).

B. at $t>0$, $V_c(t) = V(\infty) + [V(0) - V(\infty)]\,e^{-t/\tau}$, $\quad \tau = RC = 1\,s$

Capacitor is O.C, because $V(\infty) \rightarrow V_s$

$V(0) = 50V$

$\Rightarrow \quad V_c(t) = V_s + (50 - V_s)\,e^{-t/RC}$

$= 300 - 250\,e^{-t}$

C. $V_c(0.8s) = 300 - 250e^{-0.8} = 187.67\ V$

D. Energy stored in cap at 0.8s $= \frac{1}{2} C V_c(0.8s)^2 = \frac{1}{2} \times (187.67)^2 \times 1000\times10^{-6}$ J

$\simeq 17.61$ J

if good exposure ~~need~~ need at least 15J of energy stored in cap, we ~~have~~ have more than that in our cap at 0.8s → Good exposure

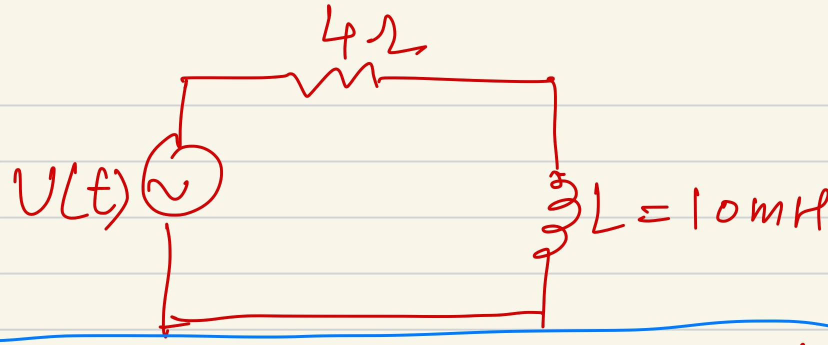
Q3. You are programming the motion control for the knee joint of a humanoid robot. The joint uses a high-performance Brushless Motor. One phase of the motor winding is modeled as a Resistor ($R = 4\ \Omega$ and an Inductor ($L = 10$ mH) in series. To hold a static crouched position, the motor driver synthesizes a steady sinusoidal voltage: *$v(t) = 24 \sin(400\ t)$ V. (Note: 400 rad/s is the electrical frequency of the drive signal).*

You need to calculate the Steady State Current flowing through the winding to ensure the motor driver chip won't overheat (Limit = 4.0 A).

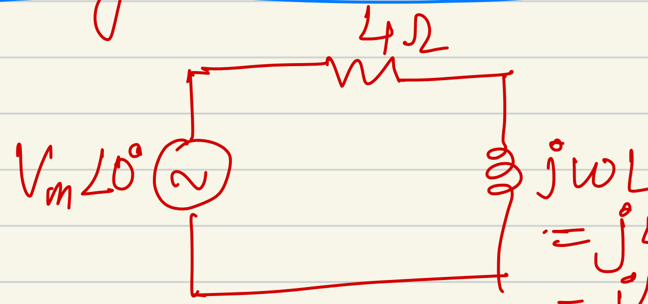
A. Write a differential equation for this circuit in the time domain to find i(t) and v(t). Why is this approach considered computationally expensive for a real-time controller?

B. Transform the problem into the Frequency Domain to solve it using algebra: convert the source v(t) to a Phasor, and calculating the Impedance (Z_{total}), find the current Phasor Magnitude and corresponding i(t).

C. Based on your Phasor result, does the current exceed the 4.0 A safety limit? Why is it safe to ignore the initial startup transient when analyzing a robot holding a pose for several seconds?



in steady state → phasor domain



$= j400 \times 10 \times 10^{-3}$

$= j4$

B. $\omega = 400$ rad/s

$V_m = 24$

$Z_{total} = 4\Omega + j4\ \Omega$

$$I = \frac{24\angle 0^\circ}{4+j4} = \frac{24\angle 0^\circ}{\sqrt{32}\ \angle \tan^{-1} 1} = \frac{24}{\sqrt{32}} \angle -45^\circ$$

$= 4.24\angle -45^\circ$

$i(t) = 4.24 \cos(400t - 45^\circ)$

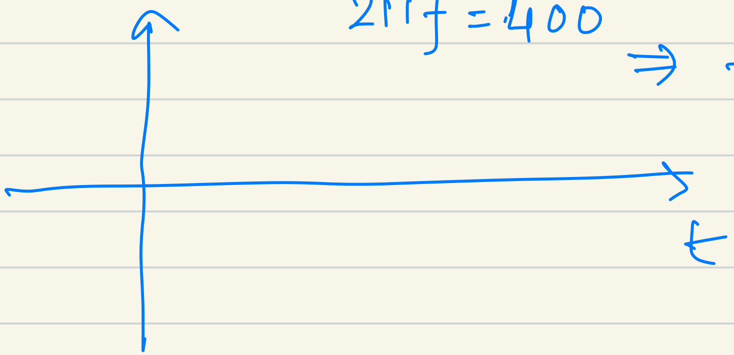
C. yes, it is exceeding 4.0A safety limit.

R-L circuit $\quad i(t) = \underbrace{\text{transient output term}}_{\propto\, e^{-R/L\, t}} + \underbrace{\text{sinusoidal term}}_{\cos(\omega t + \phi - \theta)}$

for transient, calculate $\tau = \frac{L}{R} = \frac{10\times 10^{-3}}{4}\,s = 2.5\,ms$

$2\pi f = 400 \Rightarrow f = \frac{200}{\pi} \sim 63.7\,Hz$

$T \approx \frac{1}{f} \sim 0.0157\,s \approx 15.7\,ms$



if the robot is holding the pose for several seconds → steady state response should dictate the response of the circuit.

Q4. You are building an analog Noise Filter for the robot's voice recognition microphone. The robot works in a noisy environment (humming servers and hissing steam). You need a circuit that blocks low rumble and high hiss, but passes the human voice clearly. Based on your friend's advice you use a series R-L-C circuit, where R = 100 Ω representing the Output/ADC input impedance), L = 10 mH, and C = 1 uF. Input is a microphone signal V_{in} containing many different frequencies.

A. At very low frequencies (w —> 0), how does the capacitor act like (a short circuit or an open circuit)? Will the low-frequency rumble get through to the resistor?

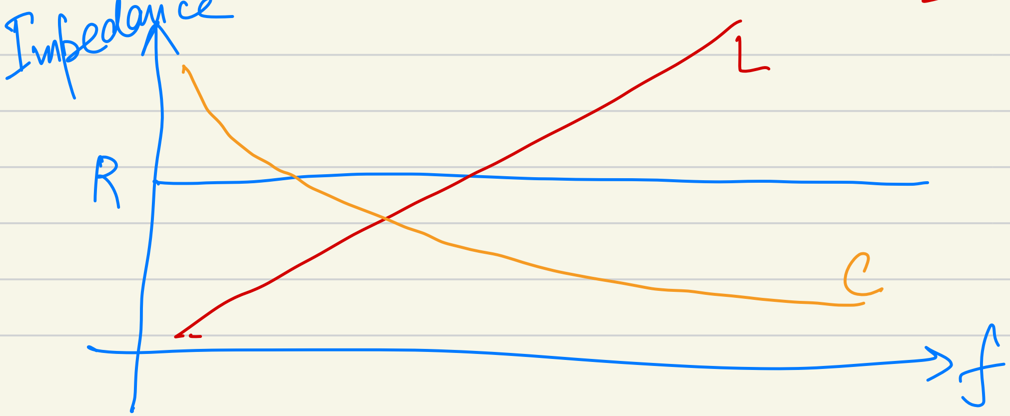
B. At very high frequencies (w —> inf), how does the inductor act like (a short circuit or an open circuit)? Will the high-frequency hiss get through?

C. Can the Inductor and Capacitor Reactances effectively cancel each other out at specific frequency? If yes, calculate that frequency.

D. Does the circuit allow maximum or minimum current at this frequency?

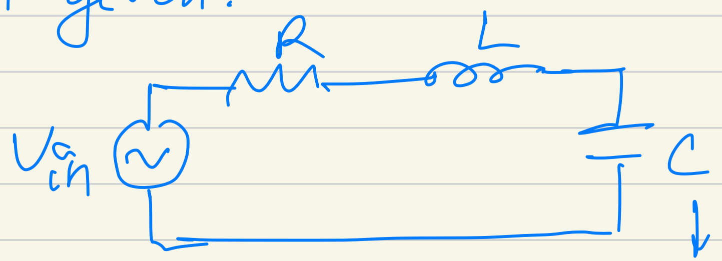
E. If you wanted to make the filter sharper (listening *only* to a specific tone and rejecting everything else aggressively), would you increase or decrease the Resistance R?

e.g. Spectrum of voice



circuit given:-

$X_L = j\omega L$



$X_C = \frac{1}{j\omega C}$

at low freq. ($\omega \to 0$)

C → O.C. (i.e DC blocker)

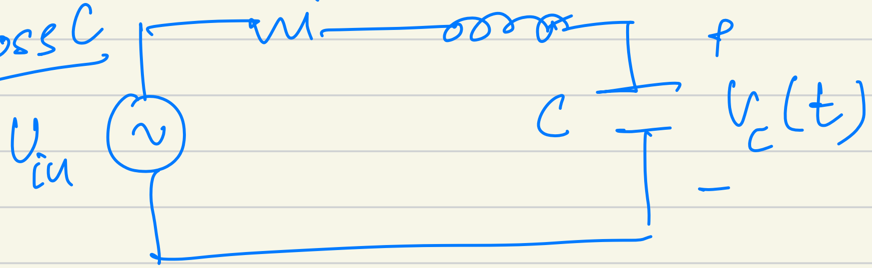
L → S.C.

at very high freq. ($\omega \to \infty$)

C → S.C.

L → O.C.

If output is taken across C



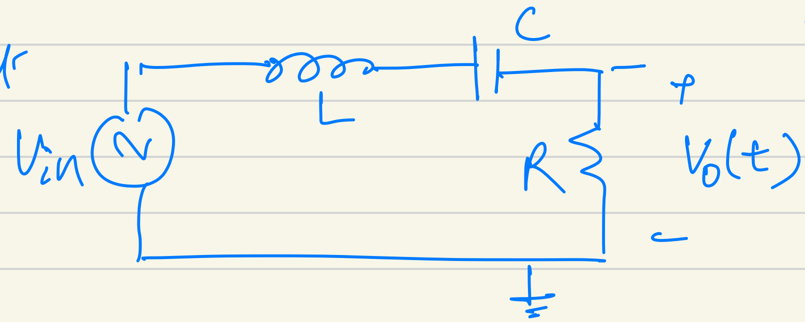
at $\omega \to 0$, $i \to 0$

$V_c(t) \Rightarrow V_{in}$

at $\omega \to \infty$, $i \to 0$ (b/c L is O.C.)

$V_c(t) \to 0$

if output is taken across R.



at $\omega \to 0$

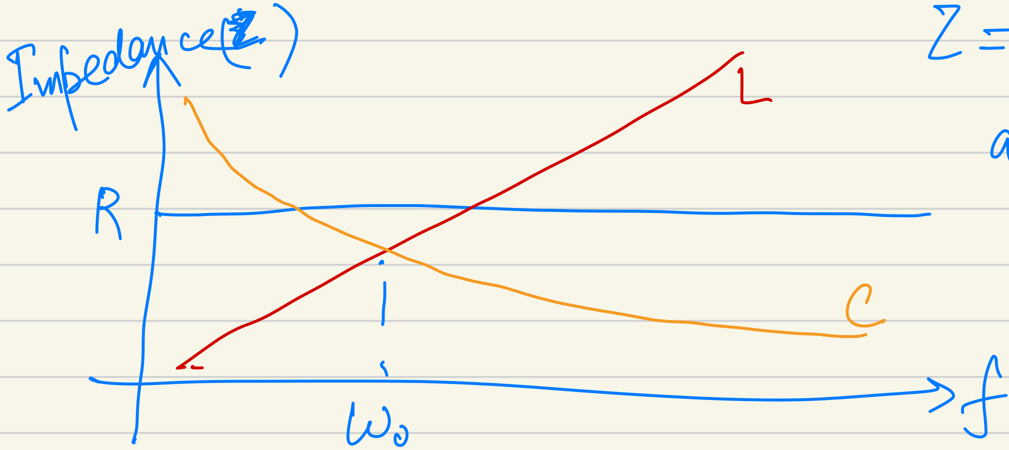
Cap is O.C.

$V_o(t) \Rightarrow 0$

at $\omega \to \infty$, $i \to 0$ (b/c L is O.C.)

$V_o(t) \to 0$

$0 < \omega < \infty$



$$Z = R + j\omega L - j\frac{1}{\omega C}$$

at ω_0, $\omega_0 L = \frac{1}{\omega_0 C}$ (i.e. $X_L = X_C$)

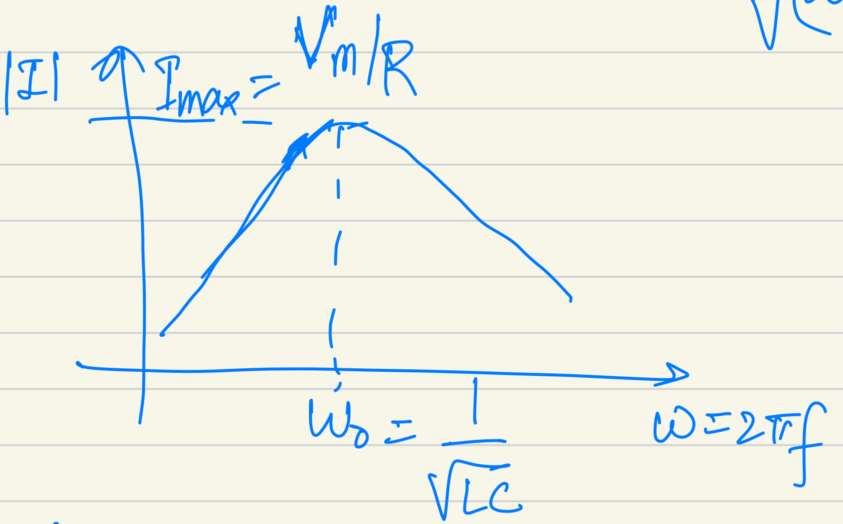
$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$Z(\omega_0) = R$$

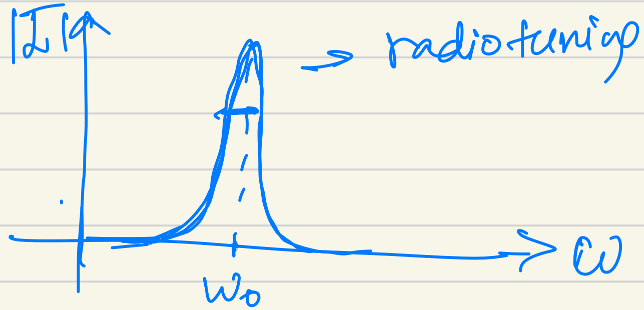
for $0<\omega<\infty$, $I = \dfrac{V_m}{\sqrt{R^2+\left(\omega L - \frac{1}{\omega C}\right)^2}\ \angle \tan^{-1}\left(\frac{\omega L - 1/\omega C}{R}\right)}$

$$|I| = \frac{V_m \cdot \omega C}{\sqrt{(\omega C R)^2 + (\omega^2 LC - 1)^2}}$$

$\omega^2 LC > 1$, $|I| < I_m$



E) desired



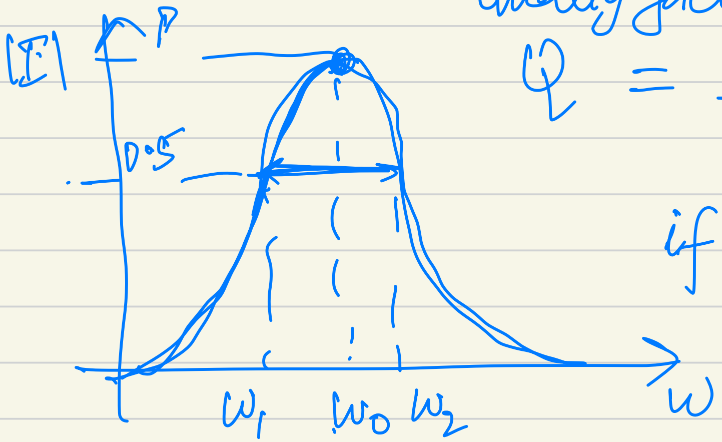
if $R \to 0$, $|I| = \dfrac{V_m \cdot \omega C}{(\omega^2 LC - 1)}$

if $\omega = \omega_0$, $|I| \to \infty$

practically → R is finite due to inductor series resistance/wire resistance.

Quality factor

$$Q = \frac{\omega_0}{\omega_1 - \omega_2}$$



if response was close to impulse $(R \to 0)$

$\omega_1 - \omega_2 \to 0$, $Q \to \infty$